

AI WORKFORCE DISPLACEMENT & TRANSFORMATION

Executive Dashboard Documentation

Documentation of the three Power BI dashboards, their business questions, executive KPIs, DAX measures, and analytical purpose.

1. Project Overview

The project analyzes how artificial intelligence is affecting workforce displacement, AI adoption, workforce transformation, and reskilling. The analysis is organized into three Power BI dashboards: AI Workforce Impact, AI Adoption & Workforce Transformation, and Workforce Adaptation & Reskilling.

2. Executive Overview — KPI Measures

The Executive Overview uses the following measures to provide a high-level summary of AI's impact on workforce conditions, adoption, adaptation, wage outcomes, and policy.

1. Avg Automation Risk

```
AVERAGE(ai_workforce_displacement[sector_automation_risk_score])
```

Purpose: Average sector automation risk score.

2. Avg Workforce Displaced

```
AVERAGE(ai_workforce_displacement[pct_sector_workforce_displaced])
```

Purpose: Average percentage of sector workforce displaced.

3. Avg New Roles Created

```
AVERAGE(ai_workforce_displacement[pct_sector_workforce_new_roles_created])
```

Purpose: Average percentage of new workforce roles created.

4. Net Workforce Change

```
AVERAGE(ai_workforce_displacement[net_workforce_change_pct])
```

Purpose: Average net workforce change percentage.

5. Total AI Layoffs

```
SUM(ai_workforce_displacement[ai_cited_layoff_announcements])
```

Purpose: Total number of AI-cited layoff announcements.

6. Avg AI Adoption

```
AVERAGE(ai_workforce_displacement[ai_adoption_index])
```

Purpose: Average AI adoption index.

7. Avg AI Tool Adoption

```
AVERAGE(ai_workforce_displacement[ai_tool_adoption_pct])
```

Purpose: Average AI tool adoption percentage.

8. Avg Wage Premium

```
AVERAGE(ai_workforce_displacement[ai_skill_wage_premium_pct])
```

Purpose: Average AI-skill wage premium percentage.

9. Avg Reskilling Programs

```
AVERAGE(ai_workforce_displacement[reskilling_programs_count])
```

Purpose: Average number of reskilling programs.

10. Avg Govt AI Policy Score

```
AVERAGE(ai_workforce_displacement[govt_ai_policy_score_1_to_10])
```

Purpose: Average government AI policy score.

3. Dashboard 1 — AI Workforce Impact

Purpose: evaluate workforce pressure created by AI through displacement, automation risk, net workforce change, and AI-cited layoffs.

Q1: Which industries have the highest workforce displacement?

Visual: Top-Left: Workforce Displacement by Industry

Analytical Focus: Compare workforce displacement across industries and identify the industries with the highest displacement.

Q2: Is higher automation risk associated with higher workforce displacement?

Visual: Top-Right: Automation Risk vs. Displacement Rate

Analytical Focus: Assess the relationship between automation risk and workforce displacement.

Q3: Which industries have positive or negative net workforce change?

Visual: Bottom-Left: Net Workforce Change by Industry

Analytical Focus: Identify industries experiencing positive versus negative net workforce change.

Q4: Which industries have the highest AI-cited layoffs?

Visual: Bottom-Right: Top 5 Industries by AI Layoffs

Analytical Focus: Rank industries by AI-cited layoff announcements and identify where AI-related layoffs are concentrated.

4. Dashboard 2 — AI Adoption & Workforce Transformation

Purpose: understand how AI adoption is evolving over time and whether adoption is associated with workforce displacement or improved workforce outcomes.

Q1: How has AI adoption changed over time?

Visual: Top-Left: AI Adoption Trend Over Time

Analytical Focus: Show the direction and change of AI adoption across the available years.

Q2: Which regions have the highest AI adoption?

Visual: Top-Right: AI Adoption by Region

Analytical Focus: Compare AI adoption across regions and identify the regions with the highest adoption.

Q3: Is higher AI adoption associated with higher workforce displacement?

Visual: Bottom-Left: AI Adoption vs Workforce Displacement

Analytical Focus: Assess the relationship between AI adoption and workforce displacement.

Q4: Is higher AI tool adoption associated with better net workforce outcomes?

Visual: Bottom-Right: AI Tool Adoption vs Net Workforce Change

Analytical Focus: Compare AI tool adoption with net workforce change to assess whether stronger tool adoption aligns with better workforce outcomes.

5. Dashboard 3 — Workforce Adaptation & Reskilling

Purpose: examine reskilling investment across industries and regions and explore its relationship with workforce outcomes, AI adoption, and wage premiums.

Q1: Does reskilling vary significantly across industries?

Visual: Top-Left: Reskilling Programs by Industry

Analytical Focus: Compare reskilling programs across industries and identify differences in reskilling activity.

Q2: Which regions invest the most in reskilling?

Visual: Top-Right: Avg Reskilling Programs by Region

Analytical Focus: Compare average reskilling programs across regions and identify the regions with the highest investment.

Q3: Is greater investment in reskilling associated with better workforce outcomes?

Visual: Bottom-Left: Reskilling Programs & Net Workforce Change by Industry

Analytical Focus: Examine the relationship between reskilling investment and net workforce change.

Q4: Which industries combine high AI adoption with high AI-skill wage premiums?

Visual: Bottom-Right: AI Adoption vs. Skill Wage Premium by Sector

Analytical Focus: Identify industries that combine strong AI adoption with high AI-skill wage premiums.

Q5: Is higher AI adoption associated with a higher AI-skill wage premium?

Use the same AI Adoption vs. Skill Wage Premium relationship to evaluate the overall association between the two measures.

6. Dataset Fields Used

Business Field	Dataset Column
Industry	industry_sector
Region	region
Year	year
AI Adoption	ai_adoption_index
Workforce Displaced	pct_sector_workforce_displaced
New Roles	pct_sector_workforce_new_roles_created
Net Workforce Change	net_workforce_change_pct
AI Layoffs	ai_cited_layoff_announcements
AI Tool Adoption	ai_tool_adoption_pct

Reskilling Programs	reskilling_programs_count
Skill Wage Premium	ai_skill_wage_premium_pct
Automation Risk	sector_automation_risk_score
Government AI Policy Score	govt_ai_policy_score_1_to_10